

# VLSI Internship Task 1 Report

## Basic Logic Gates and Half Adder Implementation

### Introduction

VLSI stands for Very Large Scale Integration. It integrates millions of transistors onto a single chip and is widely used in processors, mobile devices, memory chips, and embedded systems.

### Implemented Logic Gates

- AND Gate: Output HIGH only when all inputs are HIGH.
- OR Gate: Output HIGH if at least one input is HIGH.
- NOT Gate: Produces complement of input.
- NAND Gate: Inverse of AND gate.
- NOR Gate: Inverse of OR gate.
- XOR Gate: Output HIGH when inputs are different.

### Truth Table for Basic Gates

| A | B | AND | OR | XOR | NAND | NOR |
|---|---|-----|----|-----|------|-----|
| 0 | 0 | 0   | 0  | 0   | 1    | 1   |
| 0 | 1 | 0   | 1  | 1   | 1    | 0   |
| 1 | 0 | 0   | 1  | 1   | 1    | 0   |
| 1 | 1 | 1   | 1  | 0   | 0    | 0   |

### Half Adder Circuit

Sum = A XOR B

Carry = A AND B

| A | B | Sum | Carry |
|---|---|-----|-------|
| 0 | 0 | 0   | 0     |
| 0 | 1 | 1   | 0     |
| 1 | 0 | 1   | 0     |
| 1 | 1 | 0   | 1     |

### Observations

- All logic gates were simulated successfully.
- Outputs matched expected truth tables.
- Half Adder produced correct Sum and Carry outputs.

## **Conclusion**

Successfully implemented and verified basic logic gates and Half Adder using Logisim Evolution.